

A new compound developed in the lab was tested as a mutagen. When the Ames test was performed, the results showed the presence of many bacterial colonies following exposure. This result means the chemical _____.

- ☐ a. causes apoptosis.
- ☐ b. is radioactive.
- ☐ c. would be a great candidate to use for chemotherapy.
- ☒ d. is a mutagen and also may be a carcinogen.
- ☐ e. is safe.

DELL

The viral oncogene associated with the Rous sarcoma virus is the _____.

- ☐ a. ERBB2 gene.
- ☐ b. v-sis gene.
- ☒ c. src gene.
- ☐ d. p53 gene.
- ☐ e. RAS gene.

[Clear my choice](#)



Time left 0:03:26

Which of the following is a lipophilic primary message that crosses the plasma membrane and interacts with a cytosolic receptor?

- ☐ a. Epidermal Growth Factor
- ☐ b. TGF-beta
- ☒ c. Estrogen
- ☐ d. Pertussis toxin
- ☐ e. PDGF

[Clear my choice](#)

DELL



Time left 0:04:

Each of the following components is associated with the primary cell wall except _____.

- ☒ a. lignins.
- ☐ b. cellulose microfibrils.
- ☐ c. hemicelluloses.
- ☐ d. pectins.
- ☐ e. glycoproteins.

Clear my choice

DELL

Time left 0:05:21

Each of the following is true about Ras except

_____.

- ☐ a. Ras activity is inhibited by GTPase activating proteins (GAPs).
- ☐ b. Ras activates a cascade of phosphorylation events within the cell.
- ☒ c. Ras binds steroid hormones.
- ☐ d. Ras is monomeric.
- ☐ e. Ras is activated by Sos, a guanine nucleotide exchange factor.

[Clear my choice](#)

DELL

F5

F6

F7

F8

F9

F10

F11

F12



Each of the following is true about basal lamina except _____.

- ☒ a. it is composed primarily of fibronectin.
- ☐ b. it serves as a structural support.
- ☐ c. it can influence cell migrations in certain areas of the body.
- ☐ d. it allows for movement of some small molecules but blocks movement of large molecules.

[Clear my choice](#)



Time left 0:07

Crawling cells produce broad, thin protrusions known as _____ and pointed structures called _____ at the leading edge of the cell.

- ☐ a. filopodia; lamellipodia
- ☐ b. podosome; lamellipodia
- ☐ c. pseudopodia; focal contacts
- ☒ d. lamellipodia; filopodia

[Clear my choice](#)



DELL

Time left 0:11:

Actin bundles give shape and support to _____ found in intestinal mucosal cells.

- ☐ a. flagella
- ☐ b. nuclei
- ☒ c. microvilli
- ☐ d. lamellipodia
- ☐ e. cilia

[Clear my choice](#)

DELL

Time left 0:12:45

Compounds that inhibit receptors by preventing the natural messenger from binding are known as _____.

- ☒ a. antagonists.
- ☐ b. agonists.
- ☐ c. syndecans.
- ☐ d. arrestins.

[Clear my choice](#)

DELL

TLearn First Semester 2021

English (en)

Time left 0:13

The promotion phase of carcinogenesis is enhanced by _____.

- ☐ a. natural selection of cells exhibiting enhanced growth rate and invasive properties.
- ☐ b. a single base-pair change in genomic DNA.
- ☐ c. inactivation of Ras activity.
- ☒ d. prolonged or repeated exposure to foreign or natural agents that stimulate cell proliferation.
- ☐ e. a single exposure to a carcinogen.

[Clear my choice](#)

DELL

STLearn First Semester 2021

English (en)

Time left 0:15:58

Elastin molecules are crosslinked to each other by _____.

- ☐ a. peptide bonds between glycines.
- ☐ b. hydrophobic interactions between hydroxyprolines and glycines.
- ☐ c. none of these choices (above or below).
- ☒ d. hydrogen bonds between hydroxylysines and hydroxyprolines.
- ☐ e. covalent bonds between prolines.

Clear my choice

DELL

Time left 0:16:33

Cytoskeletal elements are found in _____.

- ☒ a. all of these choices (above and below).
- ☐ b. archaea and eukaryotes.
- ☐ c. eukaryotes.
- ☐ d. bacteria.
- ☐ e. archaea.

[Clear my choice](#)

DELL

JSTLearn First Semester 2021

English (en)

Time left 0:17:44

Each of the following is true about plant pectins except _____.

- ☐ a. pectins trap and bind water molecules.
- ☒ b. pectins are long unbranched polymers.
- ☐ c. pectins form the matrix in which cellulose microfibrils are embedded.
- ☐ d. pectins have a gel-like consistency.
- ☐ e. pectins bind adjacent cell walls together.

[Clear my choice](#)

JSTLearn First Semester 2021

English (en)

Time left 0:17:44

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[Clear my choice](#)

n on Monday 6th February at 12:30-13:32

Time left 0:18:30

Each of the following is an intracellular messenger except _____.

- ☐ a. calcium ions.
- ☐ b. cAMP.
- ☒ c. PDGF.
- ☐ d. DAG.
- ☐ e. IP3.



DELL



Time left 0:1

Each of the following is an example of microfilament-based motility and contractility except _____.

- ☐ a. cytoplasmic streaming.
- ☐ b. cytokinesis.
- ☐ c. trailing edge retraction in a moving cell.
- ☒ d. ciliary motility.
- ☐ e. pseudopodia formation.

[Clear my choice](#)

DELL



USTLearn First Semester 2021

English (en)

Time left 0:23:28

Children with hereditary retinoblastoma inherit a chromosome 13 from one parent that has which of the following defects in the RB gene?

- ☐ a. duplication
- ☐ b. inversion
- ☐ c. insertion
- ☐ d. point mutation
- ☒ e. deletion

[Clear my choice](#)

DELL

Time left 0:2

Transforming growth factor beta receptor is a _____.

- ☒ a. Receptor serine-threonine kinase
- ☐ b. Non receptor serine/threonine kinase
- ☐ c. G-protein coupled receptor
- ☐ d. Receptor tyrosine kinase

[Clear my choice](#)



DELL

F6 F7 F8 F9 F10 F11

Time left 0

Which of the following would be correctly described as juxtacrine signaling?

- ☒ a. a signal that requires physical contact between cells
- ☐ b. a signal that acts on the cell that produces it
- ☐ c. a signal that diffuses over long distances from the producing cell
- ☐ d. a signal that diffuses over short distances from the producing cell
- ☐ e. a hormonal signal

DELL

TLearn First Semester 2021

English (en) ▼

Time left 0:29:

N-CAM allows interactions between neural cells and is a member of which of the following protein families?

- ☐ a. lectins
- ☐ b. catenins
- ☐ c. selectins
- ☒ d. immunoglobulin superfamily (IgSF) proteins
- ☐ e. cadherins

[Clear my choice](#)

DELL

Time left 0

Microfilaments function in _____.

- ☐ a. maintenance of animal cell shape.
- ☐ b. cell division.
- ☐ c. cytoplasmic streaming.
- ☒ d. all of these (above and below).
- ☐ e. muscle contraction.

[Clear my choice](#)



DELL

Most hereditary forms of breast and ovarian cancer arise in women who inherit a mutant copy of either the BRCA1 or BRCA2 gene, each of which encodes a proteins involved directly in _____.

- ☐ a. inhibiting apoptosis.
- ☐ b. a spindle assembly checkpoint.
- ☒ c. repairing DNA double-strand breaks.
- ☐ d. passing through the G1 restriction point.
- ☐ e. activating Rho proteins to increase cell motility.

DELL

Time left 0:3

Oncogenes encode proteins in each of the following categories except _____.

- ☐ a. growth factors.
- ☐ b. nonreceptor protein kinases.
- ☒ c. cytochrome P450 enzymes.
- ☐ d. transcription factors.
- ☐ e. growth-factor receptors.

[Clear my choice](#)

DELL

In the axoneme, dynein sidearms are anchored on _____.

- ☐ a. B tubules.
- ☐ b. interdouplet nexin.
- ☒ c. A tubules.
- ☐ d. central pair tubules.
- ☐ e. radial spokes.

[Clear my choice](#)

DELL

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[Clear my choice](#)

DELL

Time left 0:3

CapZ _____.

- ☐ a. connects microfilaments to microtubules.
- ☐ b. causes microfilaments to lengthen by assembly at their plus ends.
- ☐ c. forms microfilament branches.
- ☐ d. depolymerizes microfilaments at their plus ends.
- ☒ e. prevents assembly at microfilament plus ends.

[Clear my choice](#)

DELL



You

2/8/21, 2:10 PM



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STLearn First Semester 2021

English (en)

Time left 0:41:

Which of the following statements is correct with respect to flagella and cilia?

- ☐ a. Cilia often have a helical pattern of movement.
- ☐ b. Cilia generally are longer than flagella.
- ☒ c. Cilia generate a force parallel to the cell surface.
- ☐ d. The flagellum is usually at the leading end of a swimming sperm cell.
- ☐ e. There usually are fewer cilia than flagella.

[Clear my choice](#)

DELL



You

2/8/21, 2:09 PM



Time left 0:42

Which of the following proteins attaches a crawling cell to the substrate in a focal adhesion?

- ☒ a. integrin
- ☐ b. actin
- ☐ c. myosins
- ☐ d. alpha-actinin

[Clear my choice](#)

DELL

Question **12**Not yet
answeredMarked out of
1.00

Flag question

The _____ is a complex network of interconnected filaments and tubules that extends throughout the cytosol from the nucleus to the inner surface of the plasma membrane.

- ☒ a. cytoskeleton
- ☐ b. desmin sheath
- ☐ c. microfilament array
- ☐ d. axoneme
- ☐ e. microtubule apparatus

[Clear my choice](#)

Time left 0:4

Striated muscle has light and dark bands. The light bands are called _____, whereas the dark bands are called _____.

- ☐ a. H zones; M zones
- ☐ b. Z lines; M lines
- ☐ c. M lines; A bands
- ☒ d. I bands; A bands
- ☐ e. A bands; I bands

[Clear my choice](#)

DELL

Time left 0:44:22

In cartilage, proteoglycans form large complexes by attaching to _____.

- ☐ a. elastin.
- ☐ b. glycosaminoglycans.
- ☐ c. laminin.
- ☐ d. collagen.
- ☒ e. hyaluronate.

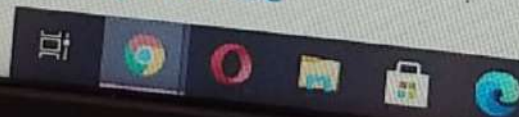
[Clear my choice](#)

DELL

How do carcinogens damage DNA?

- ☐ a. by causing breaks in one or both strands.
- ☐ b. by generating cross-links between DNA strands.
- ☒ c. all these choices (above and below) are mechanisms by which carcinogens can damage DNA.
- ☐ d. by hydroxylating DNA bases.
- ☐ e. by binding to DNA and directly disrupting base pairing.

Clear my choice



DELL



USTLearn First Semester 2021

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Engl

Time

The expression of which of the following contributes to development of cancer?

- ☒ a. oncogenes
- ☐ b. apoptosis genes
- ☐ c. proto-oncogenes
- ☐ d. tumor suppressor genes
- ☐ e. the p53 gene

[Clear my choice](#)

DELL



You

2/8/21, 2:00 PM



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JSTLearn First Semester 2021

English (en) ▾

Time left 0:51:22

Dynamic assembly and disassembly of a cytoplasmic microtubule in a cell occurs primarily at its plus end, because its minus end is usually anchored to _____.

- ☐ a. a centromere.
- ☒ b. a microtubule organizing center.
- ☐ c. the plasma membrane.
- ☐ d. a mitochondrial outer membrane.
- ☐ e. a basal body.

Clear my choice



DELL



Question **4**

Not yet
answered

Marked out of
1.00

Flag question

IP3 receptors are associated with _____.

- ☐ a. mitochondrial membranes.
- ☐ b. ER membranes.
- ☐ c. endosome membranes.
- ☒ d. plasma membranes.
- ☐ e. Golgi apparatus membranes.

[Clear my choice](#)

3

cAMP activates _____.

- ☐ a. MAPK.
- ☒ b. protein kinase A.
- ☐ c. protein kinase G.
- ☐ d. protein kinase B.
- ☐ e. protein kinase C.

[Clear my choice](#)

Which of the following forms the insoluble layer of aromatic alcohols found primarily in woody tissues?

- ☐ a. proteoglycans
- ☒ b. lignins
- ☐ c. cadherins
- ☐ d. pectins
- ☐ e. hyaluronates

[Clear my choice](#)

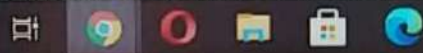


DELL



Which of the following sequences is in the correct order for the process by which a myosin head generates force on an actin filament for muscle contraction?

- ☐ a. crossbridge formation, cocking of myosin head, power stroke, crossbridge dissociation
- ☐ b. power stroke, crossbridge formation, crossbridge dissociation, cocking of myosin head
- ☐ c. crossbridge formation, power stroke, crossbridge dissociation, power stroke, cocking of myosin head



DELL

